

Sample-based Nonlinear Model Predictive Control in a Lettuce Greenhouse with Uncertain Weather Forecasting

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Abstract—Farming systems are characterized by uncertain external inputs such as weather. Operational management can benefit considerably from forecasts of these external inputs. A suitable method for management support from a control perspective is model predictive control. Input uncertainty typically propagates through a system in a nonlinear fashion and the effects on optimal control strategy, as well as performance, is still unknown for most farming applications. One method to employ uncertain forecasts for optimal control, is by means of a sample-based model predictive controller. This work presents the development of such a controller, in a case study of a lettuce greenhouse with uncertain weather forecasts. Results indicate that uncertainty has considerable impact on optimal management strategy, and on control performance.

I. INTRODUCTION

Model based control is an essential feature for management support for the operational management of farming systems [1]. Control methods enable to provide us with the machine intelligence required for full automation, or decision support. Model predictive control is a particularly useful tool since it employs information about the future of external inputs, such as weather, prices of marketable products, and availability of resources. This information is used to optimally schedule inputs such as water, fertilizer and biocides, as well as actions such as harvesting and weeding. The external inputs of farming systems are often stochastic in nature, which limits their predictability and results in uncertain forecasts. When a future external input is anticipated by use of a forecast, the uncertainty of the forecast creates a risk of under- or over-actuation of the control algorithm, and a risk of under-performance. For example, a precipitation forecast can help minimize the use of fresh water during daily field irrigation, and at the same time prevent unnecessary leaching of water, nutrients and possibly pesticides, to the ground water. However, if the forecast is uncertain, there is a risk that precipitation amounts are underestimated, resulting ultimately in yield loss due to crop water stress. A risk averse way of model predictive control could be to systematically raise the daily irrigation dosage. Such a raise will increase robustness against yield loss, but it will decrease control performance in terms of input efficiency [2].

Another example concerns greenhouse climate management. Modern greenhouses contain a wide range of climate sensors (for temperature, humidity, carbon dioxide levels, and photosynthetic radiation) as well as actuators (such as heaters, CO₂ injectors, window openers, screens for light diffusion or for extra insulation, and artificial lighting) in order to control the indoor climate. A safe management strategy is to maintain the climate at more or less steady levels during day and night, that are ideal for crop growth and development, and avoid climate variables to exceed any threshold at any time. This is robust with respect to any unforeseen fluctuations in weather, but usually not very input efficient. For example, a very high indoor humidity should be avoided since it might promote fungal diseases such as botrytis or mildew. Humidity increases due to crop transpiration, and can be decreased through ventilation by opening windows. However, ventilation usually causes loss of latent heat, which will require compensation through additional heating. Hence, low humidity levels are relatively more safe but cost are less energy efficient compared to high humidity levels. Hence, in theory a model predictive controller that is risk averse with respect to climate variables deviating too much from their set-points, can be expected to be less input efficient.

In greenhouse climate management, model predictive control that employs weather forecasts to increase input efficiency, is a well studied topic [3], [4]. An overview of studies can be found in [5]. However, the influence of the uncertainty in weather forecasts have been scarcely investigated, even though such uncertainty can have a considerable influence on climate control performance in storage rooms [6]. In [7], a robust model predictive controller is proposed that takes measurement uncertainty into account. However, from this work it was not yet clear what the impact of weather forecast uncertainty was on the control actions and performance. The impact of external input uncertainty on control actions, and control performance, is of general interest in the farming industry. Control actions need to be adapted to deal with uncertainty, and expectations on production rates should be realistic since many farmers are tied to delivery contracts.

This paper proposes a sample-based nonlinear model predictive controller, in a case study of a lettuce greenhouse, that takes into account uncertainty on the weather forecasts by propagating this uncertainty through a nonlinear model that includes greenhouse and crop dynamics. It is assumed that common statistical properties (mean, variance and probability density function) of the weather forecasts are known. The proposed controller deals with this uncertainty,

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while controlling the indoor climate at set points that are ideal for crop growth. Subsequently the effect of uncertain weather forecasts on the control actions and performance is investigated. This paper proposes:

- 1) A nonlinear controller that optimizes energy usage per kilogram crop yield, while taking uncertain weather predictions into account.
- 2) The employment of a method that allows for propagating uncertain weather forecasts through the model that is used in the controller and
- 3) it provides a contribution to the research question of how much uncertain weather forecasts influence the control results.

This paper is organized as follows. Section II presents the greenhouse and crop model that is used for simulation and testing the proposed controller. Then in Section III, the proposed sample-based controller is precised. Section IV presents the simulation results and this paper is concluded in Section V.

II. GREENHOUSE AND LETTUCE MODEL

The nonlinear model considered in this work is taken from [3] and discretized using the forward Euler method. This results in:

$$\begin{aligned} x(k+1) &= f(x(k), u(k), d(k)), \\ y(k) &= g(x(k)), \end{aligned} \quad (1)$$

with discrete time $k \in \mathbb{Z}^+$, state $x(k) \in \mathbb{R}^4$, measurement $y(k) \in \mathbb{R}^4$, controllable input $u(k) \in \mathbb{R}^3$, weather disturbance $d(k) \in \mathbb{R}^4$ and $f(\cdot), g(\cdot)$ nonlinear functions that are, together with initial conditions, given in the Appendix. The signals are defined as:

$$\begin{aligned} x(k) &= (W \quad C_{CO_2,a} \quad T_a \quad C_{H_2O,a})^T, \\ &= (x_1(k) \quad x_2(k) \quad x_3(k) \quad x_4(k))^T, \\ d(k) &= (I_o \quad C_{CO_2,o} \quad T_o \quad C_{H_2O,o})^T, \\ &= (d_1(k) \quad d_2(k) \quad d_3(k) \quad d_4(k))^T, \\ u(k) &= (u_{CO_2,o} \quad u_v \quad u_q)^T, \\ &= (u_1(k) \quad u_2(k) \quad u_3(k))^T, \end{aligned}$$

where we leave out the time dependency for ease of notation, i.e., $W = W(k), I_o = I_o(k)$ and so forth. **The state $x(k)$ contains the dry matter content of the lettuce W in $\text{kg}\cdot\text{m}^{-2}$, which is the lettuce's weight per square meter after all its water has been removed.** The dry matter content is a quantity that is commonly used to measure yield because it shows less seasonal variation with respect to fresh weight. **The state additionally contains the indoor carbon dioxide concentration $C_{CO_2,a}$ in $\text{kg}\cdot\text{m}^{-3}$, air temperature T_a in $^\circ\text{C}$ and humidity in $C_{H_2O,a}$ in $\text{kg}\cdot\text{m}^{-3}$.** The weather disturbance $d(k)$ contains the incoming radiation I_o in $\text{W}\cdot\text{m}^{-2}$ and the outside carbon dioxide concentration $C_{CO_2,o}$ in $\text{kg}\cdot\text{m}^{-3}$, temperature T_o in $^\circ\text{C}$ and humidity content $C_{H_2O,o}$ in $\text{kg}\cdot\text{m}^{-3}$. The controllable signal $u(k)$ contains the supply rate of carbon dioxide $u_{CO_2,o}$ in $\text{mg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, ventilation rate through the vents in $\text{mm}\cdot\text{s}^{-1}$

and energy supply by the heating system u_q in $\text{W}\cdot\text{m}^{-2}$. The measurement $y(k) \in \mathbb{R}^4$ contains W in $\text{g}\cdot\text{m}^{-2}$ and the indoor $C_{CO_2,a}$ in ppm, T_a in $^\circ\text{C}$ and relative humidity $C_{H_2O,a}$ in %. Note that $y(k)$ contains equivalent signals as $x(k)$, but with different (standard sensor) units for the dry weight and indoor carbon dioxide concentration and humidity. Figure 1 graphically depicts the greenhouse model with lettuce(1) and the interactions between states (outputs), disturbances and control signals. Here it can be seen that, for example, the solar radiation has a direct influence on the lettuce and indoor climate. However, **all three control signals have an indirect influence on the lettuce's dry weight, while having a direct one on the indoor climate. Hence it is via the latter that the lettuce is controlled.**

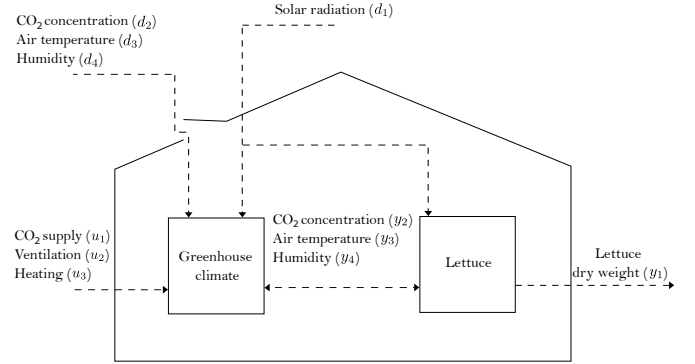


Fig. 1. Schematic representation of the greenhouse with lettuce. The arrows indicate the interaction between the greenhouse and lettuce model with the control signal $u(k)$ and meteorological disturbance $d(k)$.

The model in (1) is used as the simulation model to demonstrate the proposed controller that is the subject of the following section.

III. SAMPLE-BASED NONLINEAR MODEL PREDICTIVE CONTROLLER

The proposed controller is working following the receding horizon principle. Here, at each time step, a new state measurement is taken from the greenhouse and used to initialize the model that is used in the controller (controller model). This model is propagated forward in time from this current state measurement, while a constrained cost function is minimized with the control signals as decisions variables. From these optimized future control signals only the first is applied after which, again, a new current state measurement is collected from the greenhouse. This procedure repeats itself. So after every time step, a new state measurement is used to re-initialize the controller model. This makes it a closed-loop controller that yields, when properly tuned, a more robustly controlled greenhouse relative to the open-loop case. An additional feature of the proposed controller is the inclusion of uncertain weather forecast data. This is done by propagating multiple trajectories (samples) of the uncertain weather forecast through the controller model, while optimizing future control signals. The controller model and the optimization problem that is solved in the proposed

sample-based model predictive controller are presented in the following.

A. Controller Model

The nonlinear controller model presented in this section. It has the general form:

$$\begin{aligned}\hat{x}(k+1) &= f(\hat{x}(k), u(k), \hat{d}(k)), \\ \hat{y}(k) &= g(\hat{x}(k)).\end{aligned}\quad (2)$$

The meaning of the signals and nonlinear mappings $f(\cdot), g(\cdot)$ are equivalent as in (1). The $\hat{\cdot}$ indicates an estimation, i.e., $\hat{d}(k)$ is an estimation of $d(k)$ and so forth. The inclusion of $\hat{d}(k)$ is different from (1) where $d(k)$ is used. This is however necessary since the future evolution of $d(k)$ is not exactly known. In this work, it is proposed to model $\hat{d}(k)$ as an uncertain signal with known probabilistic properties. More precisely, it is assumed that

$$\hat{d}_j(k) \sim \mathcal{U}(\mu_{\hat{d}_j}(k), \sigma_{\hat{d}_j}(k)), \quad \text{for } j = 1, 2, 3, 4 \quad (3)$$

with uniform distribution $\mathcal{U}(\cdot)$ having a mean value $\mu_{\hat{d}_j}(k) = d_j(k)$ and variance $\sigma_{\hat{d}_j} = \frac{1}{12}(\hat{d}_{j,\max}(k) - \hat{d}_{j,\min}(k))^2$. Here

$$\hat{d}_{j,\max}(k) = (1+\gamma)d_j(k) \text{ and } \hat{d}_{j,\min}(k) = (1-\gamma)d_j(k), \quad (4)$$

assuming $d_j(k) \geq 0$ and with $0 \leq \gamma \leq 1$ a scalar parameter that indicates the size of the uncertainty. The assumption that $\hat{d}_j(k)$ is uniformly distributed is justified by the fact that this distribution has a compact support, which is realistic for relatively short term weather forecasts. The uncertain solar radiation $\hat{d}_1(k)$ is graphically illustrated in Fig. 2 as an example. Here, $d_1(k)$ is the true solar radiation that is defined as the mean value of the uncertain $\hat{d}_1(k)$. The minimum $\hat{d}_{1,\min}$ and maximum $\hat{d}_{1,\max}$ values determine the variance of the uniformly distributed solar radiation. Note that each individual signal in $\hat{d}(k)$ is modeled in this way and that still general patterns on the future weather forecasts are assumed to be known. However, the forecasts are considered as uncertain in the controller model, which is different from previous related scientific work.

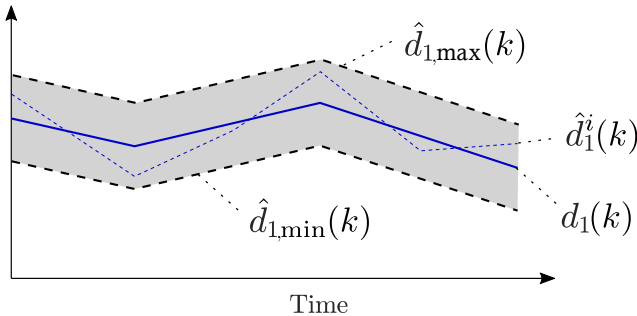


Fig. 2. Graphical illustration of the uncertain disturbance signal. The solar radiation's uncertainty $\hat{d}_1(k)$ is depicted as an example. The mean of $\hat{d}_1(k)$ is the true radiation $d_1(k)$ and its minimum and maximum are denoted as $\hat{d}_{1,\min}$ and $\hat{d}_{1,\max}$, respectively. The solar radiation takes values from the gray area.

A challenge is that each uniform distribution contains an infinite number of realizations. One such realization is shown

in Fig.2 as $\hat{d}_1^i(k)$ and all the other infinite realizations are found in the gray area. It is impossible, from a computational point of view, to take all these realizations into account with the controller that is proposed in this work. One solution is to consider only the worst cases with the controller (for example, $\hat{d}_{1,\min}(k)$ and $\hat{d}_{1,\max}(k)$). However, it is not given that a maximum disturbance value results in a maximal model output $\hat{y}(k)$ due to the fact that $f(\cdot), g(\cdot)$ are nonlinear mappings. Another option is to draw N_s random samples (trajectories) from each uniform distribution and then take all these trajectories into account with the controller. This approach is followed in this work as is shown in the following section.

B. Optimization Problem

The optimization problem that is formulated in this section is employing the model given in (2). It is assumed that at each time instant, the state $x(k)$ can be measured, i.e., $\hat{x}(k_0) = x(k_0)$. Then, the following optimization problem is solved at each time step k_0 :

$$\begin{aligned}\min_{u(k)} \quad & \sum_{i=1}^{N_s} \sum_{k=k_0}^{k_0+N_p} V(u(k), \hat{y}^i(k)), \\ \text{s.t.} \quad & (2), \quad \hat{x}(k_0) = \hat{x}_0 \\ & u_{\min} \leq u(k) \leq u_{\max}, \\ & |u(k) - u(k-1)| \leq \delta u, \\ & \hat{y}_{\min}(k) \leq \hat{y}^i(k) \leq \hat{y}_{\max}(k), \text{ for } i = 1, \dots, N_s, \\ & \text{and for } k = k_0, \dots, k_0 + N_p,\end{aligned}\quad (5)$$

with $N_p \in \mathbb{R}$ the prediction horizon that determines how far the model (2) is propagated forward in time and corresponding optimized future control signals are found. This value can be seen as a tuning variable of the proposed model predictive controller. The value $N_s \in \mathbb{R}$ represents the number of realizations taken from the uniform distribution where the uncertain $\hat{d}(k)$ is assumed to live in. In Fig. 2, one such realization was depicted as $\hat{d}_1^i(k)$. So the controller takes N_s future trajectories of $\hat{d}^i(k)$ for $i = 1, \dots, N_s$ into account by propagating the controller model forward in time for each of these realizations, while finding only one set of optimized control signals accordingly. Clearly, when selecting N_s , a trade-off has to be made between computational time and the number of trajectories that are taken into account by the controller.

The cost function $V(u(k), \hat{y}^i(k))$ is defined as:

$$V(u(k), \hat{y}^i(k)) = -q_{\hat{y}_1} \cdot \hat{y}_1^i(N_p) + \sum_{j=1}^3 q_{u_j} \cdot u_j(k), \quad (6)$$

with $q_{\hat{y}_1}, q_{u_j} \in \mathbb{R}$ defined as weights in the optimization that can also be seen as tuning variables. In other words, the constrained optimization aims at finding a trade-off between the maximization of yield per square meter and the minimization of control signals (energy usage). Indeed, this trade-off can be tuned via $q_{\hat{y}_1}, q_{u_j}$.

The constraints in (5) are defined:

$$\begin{aligned} u_{\min} &= (0 \ 0 \ 0)^T, \quad \delta u = \frac{1}{10} u_{\max}, \\ u_{\max} &= (1.2 \ 7.5 \ 150)^T, \\ \hat{y}_{\min}(k) &= (0 \ 0 \ f_{\hat{y}_{3,\min}}(k) \ 0)^T, \\ \hat{y}_{\max}(k) &= (\infty \ 1.6 \ f_{\hat{y}_{3,\max}}(k) \ 70)^T, \end{aligned} \quad (7)$$

with lower and upper bound on the control signal defined by $u_{\min}, u_{\max} \in \mathbb{R}^3$, respectively, and the bound on the change of the control signal defined by $\delta u \in \mathbb{R}^3$. These bounds correspond to current practice. The time-varying lower and upper bound on the output are $\hat{y}_{\min}(k), \hat{y}_{\max}(k) \in \mathbb{R}^4$. In fact, only the third element in each of these bounds is time-varying and defined as:

$$\begin{aligned} f_{\hat{y}_{3,\min}}(k) &= \begin{cases} 15, & \text{if } d_1(k_0) < 10 \\ 20, & \text{otherwise} \end{cases}, \\ f_{\hat{y}_{3,\max}}(k) &= \begin{cases} 20, & \text{if } d_1(k_0) < 10 \\ 25, & \text{otherwise} \end{cases}. \end{aligned} \quad (8)$$

These time-varying constraints on the indoor temperature are set such that it is during the night colder than during the day in the greenhouse. This is according to findings reported in [8]. Here it is shown that lower greenhouse temperatures can later on be compensated by higher ones as long as a daily average greenhouse temperature is satisfied. A consequence of the time-varying temperature constraint is that during the night, when the outside temperature is relatively low, the controller is not working that hard to keep the temperature at a relatively high level. Since, if there is any, the sun is heating up the greenhouse during the day, less additional heating is necessary during this period to achieve relatively high temperatures in the greenhouse hence energy can be saved. The time-varying constraint on the indoor temperature is graphically illustrated in Fig. 3. Here, the gray area indicates the region where the controller model's output $\hat{y}_3^i(k)$ is controlled to.

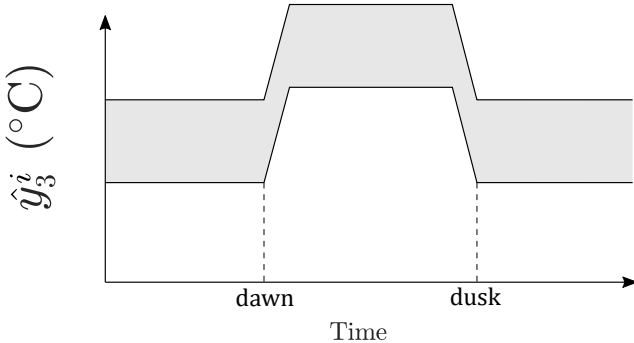


Fig. 3. Graphical illustration of the time-varying constraint that is imposed on the model's output $\hat{y}_3^i(k)$ that represents the temperature inside of the greenhouse. The gray area is the region where the optimized control signals steer $\hat{y}_3^i(k)$ in.

Note that for the time-varying constraint on the model's output it is assumed that the radiation at k_0 is known. This is

perfectly valid since such measurement devices are available and installed in greenhouses.

This ends the controller model and sample-based controller formulation. In the following, simulation results are presented.

IV. SIMULATION RESULTS

The weather data $d(k)$ used throughout the simulations is real-life data that is presented in [9]. This data is collected during experiments performed in the greenhouse called "the Venlow Energy greenhouse" that is located in Bleiswijk, Holland. The collected data points are sampled at 5 minutes and N of these are used and re-sampled to the sample period h . Other settings that are used during the simulation studies are given in Table I. The weights $q_{\hat{y}_1}, q_{u_i}$ are tuned such that an acceptable trade-off between yield and energy usage is acquired. The prediction horizon N_p is not taken too large to prevent the necessity of including uncertainty that grows over time. Indeed, weather forecasts become more uncertain over the future horizon.

TABLE I

SIMULATION AND CONTROLLER SETTINGS.

parameter	value	parameter	value
h	15 minutes	$q_{\hat{y}_1}$	10^3
N_p	6 hours	q_{u_i}	$\{10, 1, 1\}$
N	3840 (40 days)	N_s	10

The open-source software CasADi [10] and solver IPOPT [11] are used in a Matlab environment to solve the optimization problem formulated in (5), while following the direct single-shooting method and warm start option of IPOPT. Figure 4 shows the first two days of the deterministic simulation case as an example where $\hat{d}(k) = d(k)$ and $N_s = 1$. The measurement $y(k)$ is on the top, weather disturbance $d(k)$ in the middle and control signal $u(k)$ in the bottom row. Upper and lower bounds on the signals are indicated in red and black dashed, respectively. It can be seen that the controller injects additional CO_2 (u_1) in the greenhouse during the day (when there is radiation). This is due to the fact that during these periods photosynthesis is taking place and consequently CO_2 is consumed and the dry weight y_1 increases. Observe also that the bounds on the indoor temperature (y_3) are respected mainly through the utilization of heating (u_3).

Two measures are introduced to compare an uncertain case with the deterministic case. The first is a relative change of the dry weight, i.e.,

$$\delta W = \left(\frac{y_{1,\hat{d}}(N)}{y_{1,d}(N)} - 1 \right) \cdot 100 \%, \quad (9)$$

with N the total number of samples in the simulation, $y_{1,d}(k)$ the dry weight of the deterministic case and $y_{1,\hat{d}}(k)$ of the uncertain case with uncertainty level defined by γ (see (4)). The second is a relative change in rms value of the control signals, i.e.,

$$\delta \text{rms}(u_j) = \left(\frac{\text{rms}(u_{j,\hat{d}}(k))}{\text{rms}(u_{j,d}(k))} - 1 \right) \cdot 100 \%, \quad (10)$$

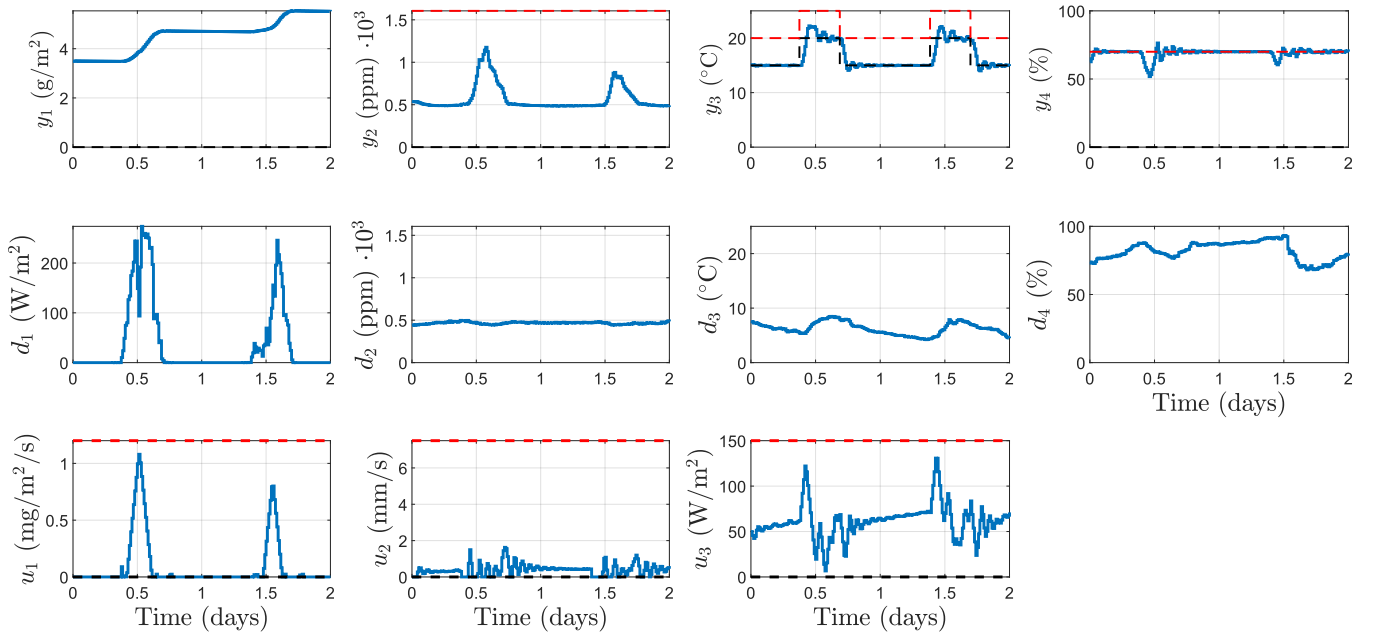


Fig. 4. First two days of the deterministic simulation results with measurements $y(k)$ in the top, weather disturbance $d(k)$ in the middle and control signal $u(k)$ in the bottom row. Upper and lower bounds on the signals are indicated in red and black dashed, respectively.

for $j = 1, 2, 3$ and $u_{j,d}(k)$ the j^{th} control signal applied in the deterministic case and $u_{j,\hat{d}}(k)$ the control signal applied in the uncertain case. Indeed, a positive $\delta\text{rms}(u_j)$ indicates that more energy is used throughout the uncertain case compared to the deterministic case. Multiple simulation are performed with each a different level of uncertainty (γ). Each of these uncertain cases is compared with the deterministic simulation case by using (9) and (10). The results are shown in Table II.

TABLE II
SIMULATION RESULTS: PERCENTAGES ARE WITH RESPECT TO THE DETERMINISTIC CASE WHERE $\hat{d}(k) = d(k)$ IS ASSUMED.

γ	δW %	$\delta\text{rms}(u_1)$ %	$\delta\text{rms}(u_2)$ %	$\delta\text{rms}(u_3)$ %
0.05	-0.63	-1.34	6.83	2.30
0.1	-1.13	-1.54	13.76	5.43
0.15	-1.83	-1.43	19.03	9.20
0.2	-2.63	-1.09	22.49	13.62
0.25	-3.56	-1.28	26.25	18.55
0.3	-4.95	-3.74	29.11	23.28

The general trend is that when the weather forecast's uncertainty increases, the dry weight yield decreases and the energy usage increases except for $u_1(k)$, the CO_2 injection. Note that a 5% decrease in dry weight is, when assuming that 96% of a fresh lettuce is water, a bit more than half a lettuce less per square meter. This is for relatively large greenhouses significant. Also, the rms values of $u_2(k)$ and $u_3(k)$ (ventilation and heating) increase significantly when the level of uncertainty in the weather forecasts increases. This implies that more energy is needed to optimally control the greenhouse climate and development of the lettuce. Interestingly, the rms value of $u_1(k)$ decreases when uncertainty

in the weather forecasts increases. A possible explanation for this is that an uncertainty increase results in more ventilation (see $\delta\text{rms}(u_2)$ column) and consequently, injecting more CO_2 is not sufficiently effective since it will exit the greenhouse via ventilation.

The results indicate that, when a grower want to invest in a predictive controller, uncertain weather forecasts need to be taken into account for having a better financial overview of the investment since these affect the energy usage per kilogram yield. Certainly because there is always a level of uncertainty on the weather forecasts.

V. CONCLUSIONS

The results indicate a clear trade-off between robustness against uncertain external inputs, and optimality in the case of indoor climate control for a lettuce cultivation. Uncertainty was shown to have a large influence on control actions, as well as control performance. Although more research is required, the results indicate that a grower has to take weather forecasts uncertainty into account when he or she want to invest in a model-based predictive controller that optimizes the greenhouse climate and crop yield.

A potential extension to this work is the inclusion of uncertainty that becomes larger as time progresses, which is even more realistic than the uncertainty model proposed in this work. Indeed, forecasts further in the future become relatively more uncertain. Another research question that can be investigated with the proposed controller is the individual effect of uncertain weather components on the control results. In this work it is assumed that all four weather forecast signals are uncertain. However, it is also possible to, for example, only make the radiation prediction uncertain and investigate its consequence. The results could be used by the

supporting industry for greenhouse management in making an educated choice on what type of weather forecast to focus on or to improve upon.

Although these results should be further corroborated by more extensive case studies, the outcomes of this study suggest that uncertainty of weather forecasts forms a relevant aspect in climate management for indoor crop cultivation. Given the uncertain nature of weather forecasts, and their prominent roles in daily management of various farming systems such as field irrigation and climate control in animal housing systems, trade-offs similar to the one following from this study, can be expected. Uncertainty based model predictive control based on external input is therefore a promising topic of investigation in the context of operational farm management.

APPENDIX

The discrete greenhouse with lettuce model is defined as:

$$\begin{aligned} x_1(k+1) &= x_1(k) + h \left(c_{1,1} \phi_{\text{phot,c}}(k) - \dots c_{1,2} x_1(k) 2^{x_3(k)/10-5/2} \right), \\ x_2(k+1) &= x_2(k) + \frac{h}{c_{2,1}} \left(-\phi_{\text{phot,c}}(k) + c_{2,2} x_1(k) \dots 2^{x_3(k)/10-5/2} + u_1(k) 10^{-6} - \phi_{\text{vent,c}}(k) \right), \\ x_3(k+1) &= x_3(k) + \frac{h}{c_{3,1}} \left(u_3(k) - (c_{3,2} u_2(k) 10^{-3} + c_{3,3}) \dots (x_3(k) - d_3(k)) + c_{3,4} d_1(k) \right), \\ x_4(k+1) &= x_4(k) + \frac{h}{c_{4,1}} \left(\phi_{\text{transp,h}}(k) - \phi_{\text{vent,h}}(k) \right), \end{aligned}$$

with

$$\begin{aligned} \phi_{\text{phot,c}}(k) &= \left(1 - \exp(-c_{1,3} x_1(k)) \right) \left(c_{1,4} d_1(k) (-c_{1,5} x_3(k)^2 + c_{1,6} x_3(k) - c_{1,7}) \dots (x_2(k) - c_{1,8}) \right) / \varphi(k), \\ \varphi(k) &= c_{1,4} d_1(k) + (-c_{1,5} x_3(k)^2 + c_{1,6} x_3(k) - \dots c_{1,7}) (x_2(k) - c_{1,8}), \\ \phi_{\text{vent,c}}(k) &= (u_2(k) 10^{-3} + c_{2,3}) (x_2(k) - d_2(k)), \\ \phi_{\text{vent,h}}(k) &= (u_2(k) 10^{-3} + c_{2,3}) (x_4(k) - d_4(k)), \\ \phi_{\text{transp,h}}(k) &= c_{4,2} \left(1 - \exp(-c_{1,3} x_1(k)) \right) \dots \left(\frac{c_{4,3}}{c_{4,4} (x_3(k) + c_{4,5})} \exp\left(\frac{c_{4,6} x_3(k)}{x_3(k) + c_{4,7}}\right) - x_4(k) \right), \end{aligned}$$

with h the sample period. Here, $\phi_{\text{phot,c}}(k)$, $\phi_{\text{vent,c}}(k)$, $\phi_{\text{transp,h}}(k)$ and $\phi_{\text{vent,h}}(k)$ are the gross canopy photosynthesis rate, mass exchange of CO_2 through the vents, canopy transpiration and mass exchange of H_2O through the vents, respectively. The initial state and control signal are: $x(0) = (0.0035 \ 0.001 \ 15 \ 0.008)^T$, $u(0) = (0 \ 0 \ 50)^T$. The

measurement equation is defined as:

$$\begin{aligned} y_1(k) &= 10^3 x_1(k) & \text{g m}^{-2}, \\ y_2(k) &= \frac{c_{2,4} (x_3(k) + c_{2,5})}{c_{2,6} c_{2,7}} \cdot x_2(k), & \text{ppm} \cdot 10^3, \\ y_3(k) &= x_3(k), & ^\circ\text{C}, \\ y_4(k) &= \frac{c_{2,4} (x_3(k) + c_{2,5})}{\exp\left(\frac{c_{4,8} x_3(k)}{x_3(k) + c_{4,9}}\right)} \cdot x_4(k), & \%, \end{aligned}$$

The model parameters $c_{i,j}$ are chosen following [3] and given in Table III.

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TABLE III

VALUES OF THE MODEL PARAMETERS THAT ARE TAKEN FROM [3].

parameter	value	parameter	value	parameter	value	parameter	value
$c_{1,1}$	0.544	$c_{2,1}$	4.1	$c_{3,1}$	$3 \cdot 10^4$	$c_{4,1}$	4.1
$c_{1,2}$	$2.65 \cdot 10^{-7}$	$c_{2,2}$	$4.87 \cdot 10^{-7}$	$c_{3,2}$	1290	$c_{4,2}$	0.0036
$c_{1,3}$	53	$c_{2,3}$	$7.5 \cdot 10^{-6}$	$c_{3,3}$	6.1	$c_{4,3}$	9348
$c_{1,4}$	$3.55 \cdot 10^{-9}$	$c_{2,4}$	75560	$c_{3,4}$	0.2	$c_{4,4}$	8314
$c_{1,5}$	$5.11 \cdot 10^{-6}$	$c_{2,5}$	273.15			$c_{4,5}$	273.15
$c_{1,6}$	$2.3 \cdot 10^{-4}$	$c_{2,6}$	101325			$c_{4,6}$	17.4
$c_{1,7}$	$6.29 \cdot 10^{-4}$	$c_{2,7}$	0.044			$c_{4,7}$	239
$c_{1,8}$	$5.2 \cdot 10^{-5}$					$c_{4,8}$	17.269
						$c_{4,9}$	238.3

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